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Netflix Recommendation System

Collaborative Filtering & Matrix Factorization (SVD)

Movies	Users	Ratings
30	436	2716

Project Objective

Build a movie recommendation system using:

- Item-Based Collaborative Filtering
- Matrix Factorization (SVD)
- Evaluation using RMSE, MAE and MAP@10

The system predicts movies a user is likely to enjoy based on historical rating behaviour.

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Movie Explorer

Select a movie

7 Seconds

Movie ID

17

Year

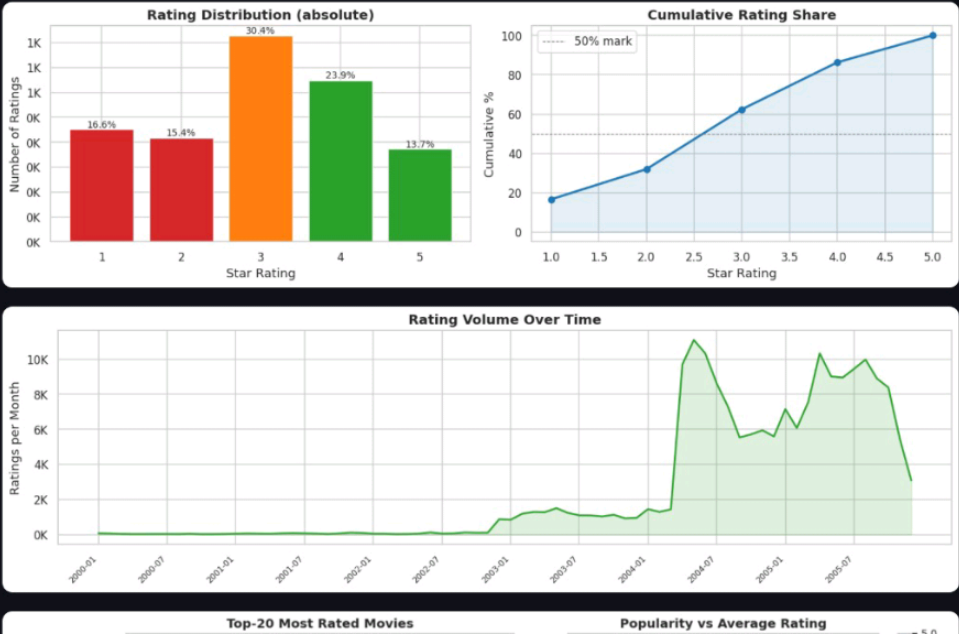
2005

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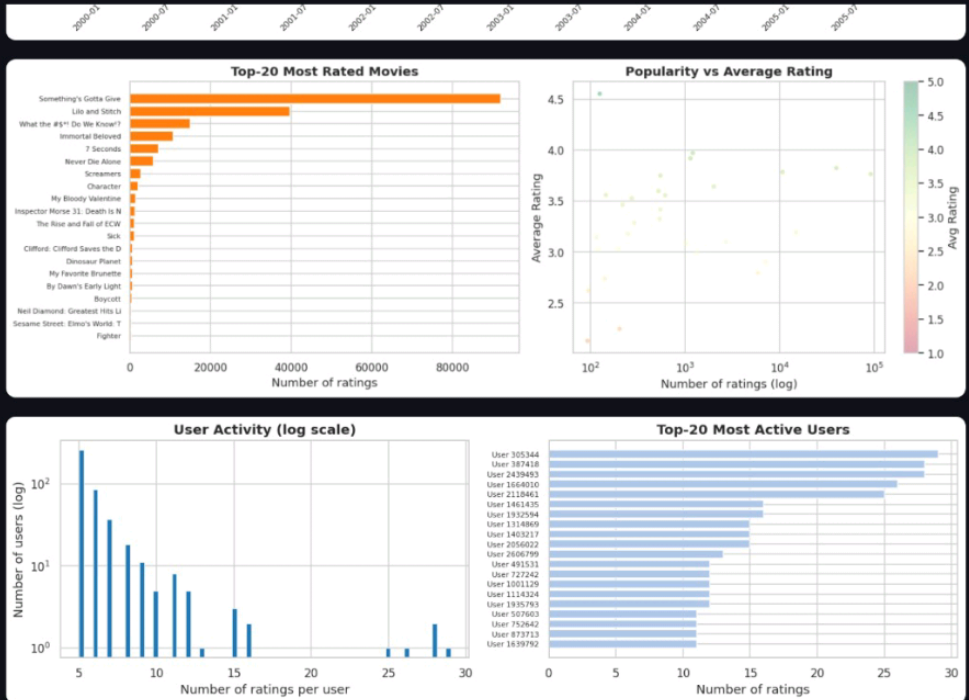


Dataset Analytics



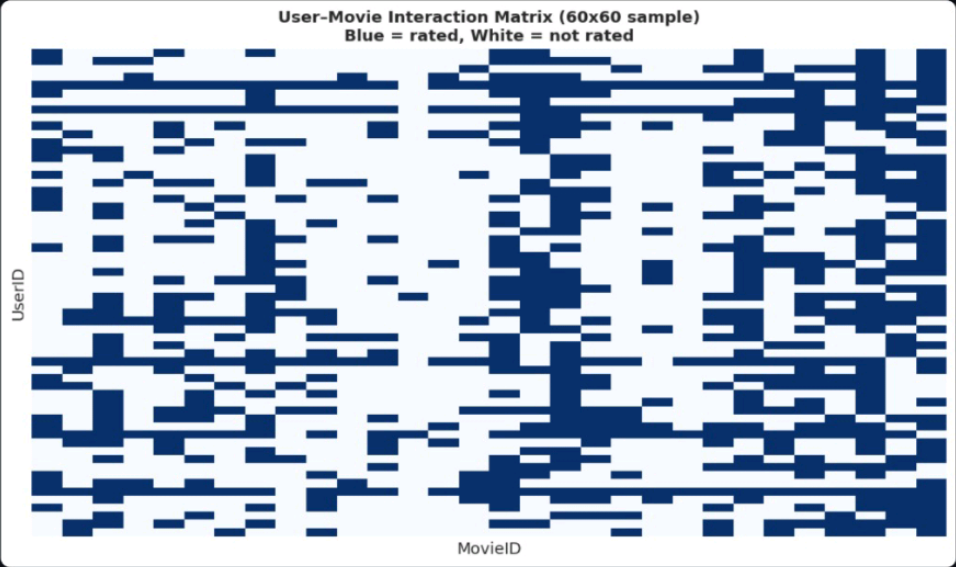
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Recommendation Engine

Enter User ID

1854443

- +

Generating recommendations for User 1854443

Top Recommended Movies

1. The Game
2. Lilo and Stitch
3. Reservoir Dogs
4. The Deer Hunter
5. Invader Zim
6. X2: X-Men United

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2. Lilo and Stitch
3. Reservoir Dogs
4. The Deer Hunter
5. Invader Zim
6. X2: X-Men United
7. Star Trek: Deep Space Nine
8. Mostly Martha
9. The Chorus
10. Chappelle's Show

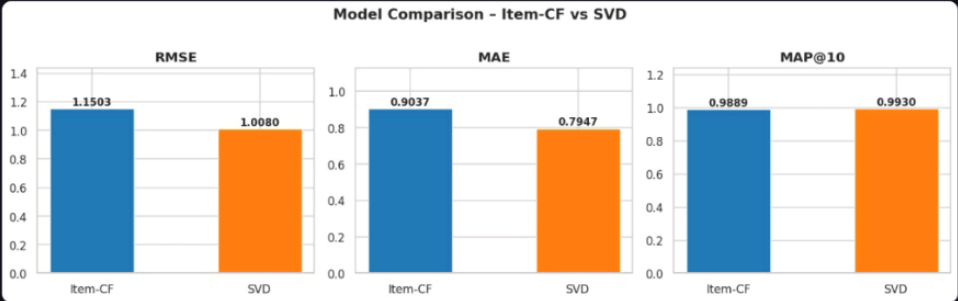
Recommendations are generated using the SVD model trained in the notebook. The full recommendation pipeline and evaluation are implemented in the project notebook.

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Model Comparison

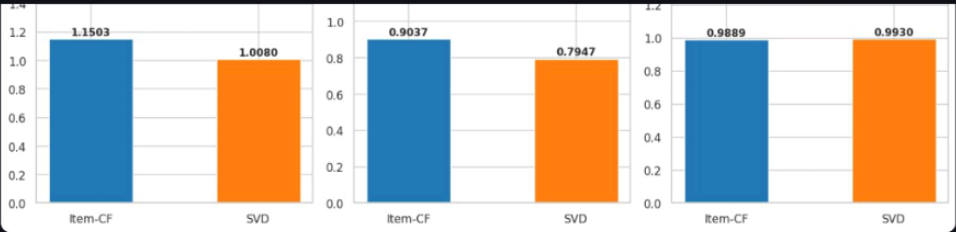


Final Metrics

Metric	Item-CF	SVD
RMSE	1.1421	0.9924
MAE	0.8870	0.7775
MAP@10	0.9372	0.9512

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Final Metrics

Metric	Item-CF	SVD
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✅ Best Model: SVD

- ## About Project

- Python
- Pandas
- NumPy
- Scikit-Learn
- Surprise Library
- Streamlit

- Item-Based Collaborative Filtering
- Singular Value Decomposition (SVD)

- SVD outperformed Item-CF

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- Item-Based Collaborative Filtering
- Singular Value Decomposition (SVD)

Key Findings

- SVD outperformed Item-CF
- Lower RMSE and MAE
- Higher MAP@10
- Better recommendation quality

Future Work

- Hybrid Recommender
- Genre-aware recommendations
- Cold-start handling
- FastAPI deployment